

Global Facilities - Design & Construction Standard - CSA - FAB, CUB, & Admin/Probe Structural Loading Specifications

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[ECN History](#)

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1.0 Purpose

- 1.1 This document serves as a System Standard and Guideline on specification that shall be followed and complied in the Design, Engineering and Construction of FAB, CUB, & Admin/Probe Building Structural Loadings as part of any green and/or brown field construction projects.
- 1.2 The standard is applicable for any green and/or brown field construction projects with regards to provide FAB, CUB, & Admin/Probe Building Structural Loadings as per required in the contract.
- 1.3 The document is intended to serve as a reference and guideline in addition to any Construction Drawings and Specification provided in the contract requirement including any required by local authority regulation and standard requirement as well as code of practice.
- 1.4 The established Specifications Standard for each locale shall also comply with Micron's in-house:
 - 1.4.1 RBA Code of Conduct requirements for Semiconductor Manufacturing
 - 1.4.2 Environmental, Health & Safety Standards
 - 1.4.3 Operational Requirements /SOP (Standard Operation Procedures)
 - 1.4.4 Security Protocol
- 1.5 The Specifications Standard to be adopted in each local shall not violate any local codes or by-laws and reflects the current semi-conductor industry's best practices.
- 1.6 Where there is the option to adopt either the MasterSpec or in-region specifications standard, the AE (Architectural & Engineering Consultants) shall

propose the most suitable solution based on safety, quality, time & cost to Micron's Designated Project Team (DPT) for review and acceptance. Otherwise, the more stringent option shall apply.

2.0 Scope

- 2.1 This document for FAB, CUB, Admin/Probe Building structural loading specifications covers all necessary related design, engineering, procurement, construction, installation and completion of piling, substructure, superstructure, and roof structure to full compliance of the contract requirement and required for the project. It extended to cover any green and/or brown field construction projects.
- 2.2 This document applies to all Micron Facilities employees, contractors, and vendors at all Micron sites.

3.0 Definitions and Acronyms

- 3.1 Definitions of acronyms used in this document are listed as follows:

Term/Acronym	Definition / Description
AE	External Architectural & Engineering Consultants
DPT	Micron's Designated Project Team
EHS	Micron's Environmental Health & Safety Team
FCT	Facilities Central Team. This is used interchangeably with Global Facilities Engineering Team and associated to O&M BKMs
GFC	Global Facilities & Construction Team. This is used interchangeably with Global Facilities (GFac), Facilities Central Group (FCG), World Wide Operations (WWOps), World Wide Facilities (WW_FACILITIES), Corp Facilities
GFCT	Micron's Global Facilities Construction Team
GFTT	Global Facilities Technology Team. This is used interchangeably with Global Facilities Engineering Team and associated to design standards
Global Facilities Central Team	This is used interchangeably with Innovation & Integration Team
Global Facilities Construction Team	This is used interchangeably with Global Construction Team
Global Facilities Operations Support Team	This is used interchangeably with Global Facilities Operations Team
Global Facilities Planning Team	This is used interchangeably with Global Facilities Line Planning Team
QP/PE	External Qualified Person (professional architect/engineer)
SME	Micron's Subject Matter Expert
TM	Micron's Team Members

4.0 STRUCTURAL LOADS

This document specifies the minimum design forces including dead load, live load, wind and earthquake loads, and their various combinations. These loads shall be applicable for the design of buildings and structures in conformance with the general design requirements.

4.1 DEAD LOADS

Dead Load is the vertical load due to the weight of permanent structural and non-structural components and attachments of a building such as walls, floors, ceilings, permanent partitions, and fixed service equipment etc. Dead load for a structural member shall be assessed based on the forces due to:

- weight of the member itself,
- weight of all materials of construction incorporated into the building to be supported permanently by the member,
- weight of permanent partitions,
- weight of fixed service equipment

DEAD LOADS	
1. Specific Weight of Concrete	25 kN/m ³
2. Specific Weight of Steel	78 kN/m ³
3. Typical Floor Finishes	1.2 kN/m ²
4. 100mm Thick Brickwall	2.8 kN/m ²
5. 200mm Thick Brickwall	5 kN/m ²

4.2 LIVE LOADS

Live load is the load superimposed by the use or occupancy of the building not including the environmental loads such as wind load, earthquake load, and snow loads.

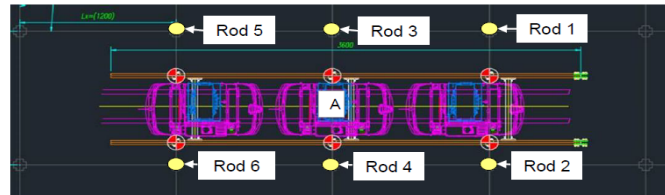
4.3 VIBRATION LOADS

Micron VC floor vibration guidelines for cleanroom areas are based on root-mean-square vibration velocity in the 1/3rd octave frequency spectra. Refer to the typical cleanroom AMHS loading below.

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In this calculation only one (1) vehicle, i.e. vehicle A is considered passing Rod 3 and Rod 4. The other two vehicles in the figure above are excluded in the calculation.

- Total Mass = 130kg (Vehicle) + 8kg (Payload) = **138kg**
- Dynamic load on Rods 3 & 4 = $138 \times 9.8 = \mathbf{1353.8N}$
- Speed of vehicle = **2.5m/s**
- Time takes to travel from 1 grid line = $1.2/4.17 = \mathbf{0.48sec}$

In this case the floor structures with vibration requirement shall be design with high dynamic stiffness. For example, to achieve VC-D+ requirements, the structural framing should be precast waffle slab. For VC-C requirements, use precast DT/ST slab. Refer to the standard vibration limits tabulation below.

Legend**Frequency, Hz**

Criterion Curve	Amplitude ¹ μm/s (μin/s)	Detail Size ² μm	Description of use
Workshop (ISO)	800 (32,000)	N/A	Distinctly perceptible vibration. Appropriate to workshops and nonsensitive areas.
Office (ISO)	400 (16,000)	N/A	Perceptible vibration. Appropriate to offices and nonsensitive areas.
Residential day (ISO)	200 (8,000)	75	Barely perceptible vibration. Appropriate to sleep areas in most instances. Usually adequate for computer equipment, hospital recovery rooms, semiconductor probe test equipment, and microscopes less than 40X.
Operating theatre (ISO)	100 (4,000)	25	Vibration not perceptible. Suitable in most instances for surgical suites, microscopes to 100X and for other equipment of low sensitivity.
VC-A	50 (2,000)	8	Adequate in most instances for optical microscopes to 400X, microbalances, optical balances, proximity and projection aligners, etc.
VC-B	25 (1,000)	3	Appropriate for inspection and lithography equipment (including steppers) to 3μm line widths.
VC-C	12.5 (500)	1 - 3	Appropriate standard for optical microscopes to 1000X, lithography and inspection equipment (including moderately sensitive electron microscopes) to 1 μm detail size, TFT-LCD stepper/scanner processes.
VC-D	6.25 (250)	0.1 - 0.3	Suitable in most instances for demanding equipment, including many electron microscopes (SEMs and TEMs) and E-Beam systems.
VC-E	3.12 (125)	< 0.1	A challenging criterion to achieve. Assumed to be adequate for the most demanding of sensitive systems including long path, laser-based, small target systems, E-Beam lithography systems working at nanometer scales, and other systems requiring extraordinary dynamic stability.
VC-F	1.56 (62.5)	N/A	Appropriate for extremely quiet research spaces; generally difficult to achieve in most instances, especially cleanrooms. Not recommended for use as a design criterion, only for evaluation.
VC-G	0.78 (31.3)	N/A	Appropriate for extremely quiet research spaces; generally difficult to achieve in most instances, especially cleanrooms. Not recommended for use as a design criterion, only for evaluation.

¹ As measured in one-third octave bands of frequency over the frequency range 8 to 80 Hz (VC-A and VC-B) or 1 to 80 Hz (VC-C through VC-G).

² The detail size refers to line width in the case of microelectronics fabrication, the particle (cell) size in the case of medical and pharmaceutical research, etc. It is not relevant to imaging associated with probe technologies, AFMs, and nanotechnology.

The information given in this table is for guidance only. In most instances, it is recommended that the advice of someone knowledgeable about applications and vibration requirements of the equipment and processes be sought.

Reprinted with permission from Colin Gordon Associates. VCA–VCG refer to accepted standards for vibration sensitive tools and instruments. The levels displayed are RMS values measured in 1/3 octave band center frequencies. 1/3 octave plots are discussed in section 1.2.3.

4.4 WIND LOADS

The effect of wind loads and equivalent horizontal force (EHF) on the building to be considered on each site location as specified by local code authority requirements.

4.5 EARTHQUAKE LOADS

The earthquake loads should be defined as per the distance of the building to the seismic fault line as specified by local code authority requirements or equivalent.

- 4.6 Provisions of LIMITATIONS for those buildings and structures subjected based on geographic site locations, or for those subject to special loading including tornadoes, earthquake, snow, etc., site-specific local code data or analysis may be required to determine the design loadings on them. In such cases, and all other cases for which loads are not specified for loading information may be obtained from local codes or authority.

5.0 FAB Building

Refer to the tabulated loadings below. Loadings may vary as per local code requirements versus Micron standard loadings and to use whichever is higher value for design.

Table 1. Level 01 Super Imposed Dead Loads

Room Occupancy or Use	Uniform (kN/m2)
1. WWT Lift Station and UPW Tank	1.2
2. HVAC Room	1.2
3. UPW Polisher Room	1.2
4. Slurry Transfer Room	1.2
5. Slurry Dispense Room	1.2
6. Bulk Solvent Room	1.2
7. Bulk Caustic Room	1.2
8. Bulk Acid Room	1.2
9. Storage Room	1.2
10.FCC Room	1.2
11. Computer Room	1.2
12. MAU Room	1.2
13. Receiving/Shipping Dock	1.2
14. Lift Lobby	1.2
15. Equipment Move-In	1.2
16. Electrical Room	1.2
17. Mechanical Room	1.2
18. All General Areas	1.2

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Table 2. Level 01 Live loads

Room Occupancy or Use	Uniform (kN/m²)
1. WWT Lift Station and UPW Tank	50
2. HVAC Room	10
3. UPW Polisher Room	10
4. Slurry Transfer Room	10
5. Slurry Dispense Room	10
6. Bulk Solvent Room	10
7. Bulk Caustic Room	10
8. Bulk Acid Room	10
9. Storage Room	10
10. FCC Room	10
11. Computer Room	10
12. MAU Room	10
13. Receiving/Shipping Dock	15
14. Lift Lobby	15
15. Equipment Move-In	15
16. Electrical Room	15
17. Mechanical Room	15
18. All General Areas	10

Table 3. Level 02 Super Imposed Dead Loads

Room Occupancy or Use	Uniform (kN/m²)
1. HVAC Room	3.5
2. Solvent Room	3.5
3. Caustic Room	3.5
4. Acid & Water Reactive Room	3.5
5. Oxidizer Room	3.5
6. HF Room	3.5
7. Slurry Dispense Room	3.5
8. Electrical & Control Room	3.5
9. Copper Sulfate Room	3.5
10. Decon Room	2.25
11. TGCM Room	3.5
12. All Clean Sub Fab Room	3.5
13. All General Areas	3.5
14. All Staircase	1.2
15. Canopies	1.0
16. All Intakes/Air Duct	1.7

Table 4. Level 02 Live loads

Room Occupancy or Use	Uniform (kN/m2)
1. HVAC Room	10
2. Solvent Room	10
3. Caustic Room	10
4. Acid & Water Reactive Room	10
5. Oxidizer Room	10
6. HF Room	10
7. Slurry Dispense Room	10
8. Electrical & Control Room	15
9. Copper Sulfate Room	10
10. Decon Room	15
11. TGCM Room	10
12. All Clean Sub Fab Room	10
13. All General Areas	10
14. All Staircase	5
15. Canopies	0.75
16. All Intakes/Air Duct	5

Table 5. Level 03 Super Imposed Dead Loads

Room Occupancy or Use	Uniform (kN/m2)
1. All Clean Room Area	3.5
2. Clean Room EUV Area	3.5
3. Decon Room	2.25
4. Lift Lobby	2.25
5. Maintenance Area	2.25
6. MAU Room	3.5
7. All Intakes/Air Duct	1.7
8. All Staircase	1.2

Table 6. Level 03 Live loads

Room Occupancy or Use	Uniform (kN/m2)
1. All Clean Room Area	25
2. Clean Room EUV Area	35
3. Decon Room	15
4. Lift Lobby	15
5. Maintenance Area	15
6. MAU Room	15
7. All Intakes/Air Duct	5
8. All Staircase	5

Table 7. Level 04 Super Imposed Dead Loads

Room Occupancy or Use	Uniform (kN/m2)
1. Electrical Room	3.5
2. Acid Booster Room	3.5
3. Caustic Booster Room	3.5
4. Solvent Booster Room	3.5
5. Decon Room	3.5
6. Scrubber Room	3.5
7. Probe Areas	3.5
8. All General Areas	3.5
9. All Air Lock	3
10.All Staircase	1.2

Table 8. Level 04 Live loads

Room Occupancy or Use	Uniform (kN/m2)
1. Electrical Room	15
2. Acid Booster Room	10
3. Caustic Booster Room	10
4. Solvent Booster Room	10
5. Decon Room	15
6. Scrubber Room	15
7. Probe Areas	15
8. All General Areas	10
9. All Air Lock	5
10.All Staircase	5

Table 9. Level 05 Super Imposed Dead Loads

Room Occupancy or Use	Uniform (kN/m2)
1. Electrical Room	3.5
2. PCW Room	3.5
3. PV Room	3.5
4. DUPS Room	3.5
5. All General Areas	3.5
6. All Air Lock	3
7. All Staircase	1.2

Table 10. Level 05 Live loads

Room Occupancy or Use	Uniform (kN/m²)
1. Electrical Room	15
2. PCW Room	15
3. PV Room	15
4. DUPS Room	15
5. All General Areas	15
6. All Air Lock	5
7. All Staircase	5

Table 11. Level 06 Super Imposed Dead Loads

Room Occupancy or Use	Uniform (kN/m²)
1. HVAC Room	3.5
2. Electrical & Control Room	3.5
3. Solvent Room	3.5
4. DUPS Room	3.5
5. Caustic Room	3.5
6. Acid & Water Reactive Room	3.5
7. Oxidizer Room	3.5
8. HF Room	3.5
9. Slurry Dispense Room	3.5
10. Copper Sulfate Room	3.5
11. All Clean Sub Fab Room	3.5
12. All General Areas	3.5
13. All Staircase	1.2
14. All Intakes/Air Duct	1.7

Table 12. Level 06 Live loads

Room Occupancy or Use	Uniform (kN/m²)
1. HVAC Room	10
2. Electrical & Control Room	10
3. Solvent Room	10
4. DUPS Room	10
5. Caustic Room	10
6. Acid & Water Reactive Room	10
7. Oxidizer Room	10
8. HF Room	10
9. Slurry Dispense Room	10
10. Copper Sulfate Room	10

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Room Occupancy or Use	Uniform (kN/m ²)
11. All Clean Sub Fab Room	10
12. All General Areas	10
13. All Staircase	5
14. All Intakes/Air Duct	5

Table 13. Level 07 Super Imposed Dead Loads

Room Occupancy or Use	Uniform (kN/m ²)
1. All Clean Room Areas	3.5
2. Clean Room EUV Area	3.5
3. Decon Room	2.25
4. Lift Lobby	2.25
5. Maintenance Area	2.25
6. MAU Room	3.5
7. All Intakes/Air Duct	1.7
8. All Staircase	1.2

Table 14. Level 07 Live loads

Room Occupancy or Use	Uniform (kN/m ²)
1. All Clean Room Areas	25
2. Clean Room EUV Area	35
3. Decon Room	15
4. Lift Lobby	15
5. Maintenance Area	15
6. MAU Room	15
7. All Intakes/Air Duct	5
8. All Staircase	5

Table 15. Level 08 Super Imposed Dead Loads

Room Occupancy or Use	Uniform (kN/m ²)
1. Plenum Level	2.55
2. PV Room	3
3. Lift Lobby	3
4. PCW Room	3
5. All Staircase	1.2
6. Lift Roof	3.6

Table 16. Level 08 Live loads

Room Occupancy or Use	Uniform (kN/m ²)
1. Plenum Level	0.25
2. PV Room	15
3. Lift Lobby	15
4. PCW Room	15
5. All Staircase	5
6. Lift Roof	1.5

Table 17. Roof Level Super Imposed Dead Loads

Room Occupancy or Use	Uniform (kN/m ²)
1. Roof Level	1.75
2. SMS Roof Level	3.6
3. Acid/Alkali/Solvent/GHG/General Exhaust	3.6
4. Staircase Roof	3.6

Table 18. Roof Level Live loads

Room Occupancy or Use	Uniform (kN/m ²)
1. Roof Level	0.75
2. SMS Roof Level	10
3. Acid/Alkali/Solvent/GHG/General Exhaust	10
4. Staircase Roof	1.5

6.0 CUB Building

Refer to the tabulated loadings below. Loadings may vary as per local code requirements versus Micron standard loadings and to use whichever is higher value for design.

Table 1. Basement Level Super Imposed Dead Loads

Room Occupancy or Use	Uniform (kN/m ²)
1. WW Lift Station	3
2. WWT & Reclaim Room	3
3. Bio System Room	3
4. Waste Chem Lift Station	3

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Room Occupancy or Use	Uniform (kN/m ²)
5. Process Water Tank	As per Tanks Size(volume)
6. All Basement General Areas	3
7. All Staircase	1.2

Table 2. Basement Level Live loads

Room Occupancy or Use	Uniform (kN/m ²)
1. WW Lift Station	50
2. WWT & Reclaim Room	50
3. Bio System Room	50
4. Waste Chem Lift Station	50
5. Process Water Tank	0
6. All Basement General Areas	50
7. All Staircase	5

Table 3. Level 01 Super Imposed Dead Loads

Room Occupancy or Use	Uniform (kN/m ²)
1. Water Solvent Lift Station	3
2. UPW Room	3
3. Electrical/ Switch Room	3
4. All Gas Rooms	3
5. Lift Lobby	3
6. All General Areas	3
7. All Staircase	1.2

Table 4. Level 01 Live loads

Room Occupancy or Use	Uniform (kN/m ²)
1. Water Solvent Lift Station	20
2. UPW Room	20
3. Electrical/ Switch Room	30
4. All Gas Rooms	20
5. Lift Lobby	20
6. All General Areas	20
7. All Staircase	5

Table 5. Level 02 Super Imposed Dead Loads

Room Occupancy or Use	Uniform (kN/m2)
1. Chilled Water	2.5
2. HVAC Room	2.5
3. Electrical Room	2.5
4. MAU Room	2.5
5. All Gas Rooms	2.5
6. Lift Lobby	2.5
7. All General Areas	2.5
8. All Staircase	1.2

Table 6. Level 02 Live loads

Room Occupancy or Use	Uniform (kN/m2)
1. Chilled Water	30
2. HVAC Room	10
3. Electrical Room	15
4. MAU Room	10
5. All Gas Rooms	30
6. Lift Lobby	10
7. All General Areas	15
8. All Staircase	5

Table 7. Roof Level Super Imposed Dead Loads

Room Occupancy or Use	Uniform (kN/m2)
1. Cooling Tower	2.5
2. All Roof General Areas	3.6
3. Lift Roof	3.6
4. All Staircase Roof	3.6

Table 8. Roof Level Live loads

Room Occupancy or Use	Uniform (kN/m2)
1. Cooling Tower	10
2. All Roof General Areas	10
3. Lift Roof	1.5
4. All Staircase Roof	1.5

7.0 Admin/Probe Building

Refer to the tabulated loadings below. Loadings may vary as per local code requirements versus Micron standard loadings and to use whichever is higher value for design.

Table 1. Level 01 Super Imposed Dead Loads

Room Occupancy or Use	Uniform (kN/m ²)
1. Computer Room	2.2
2. Transformer Room	2.2
3. LV Switch Room	2.2
4. All Meeting/Interview Room	2.2
5. All Toilet Room	2.2
6. All Corridor Room	2.2
7. Public Lobby Room	2.2
8. Receiving/Shipping Dock	2.2
9. Service Lift Lobby	2.2
10. Probe Areas	2
11. Electrical Room	2.2
12. Mechanical Room	2.2
13. All General Areas	2.2

Table 2. Level 01 Live loads

Room Occupancy or Use	Uniform (kN/m ²)
1. Computer Room	10
2. Transformer Room	10
3. LV Switch Room	10
4. All Meeting/Interview Room	5
5. All Toilet Room	5
6. All Corridor Room	5
7. Public Lobby Room	5
8. Receiving/Shipping Dock	10
9. Service Lift Lobby	10
10. Probe Areas	10
11. Electrical Room	10
12. Mechanical Room	10
13. All General Areas	5

Table 3. Level 02 & Above Super Imposed Dead Loads

Room Occupancy or Use	Uniform (kN/m2)
1. Service Lift Lobby	2.7
2. Probe Areas	2.7
3. Electrical Room	2.7
4. Mechanical Room	2.7
5. All General Areas	2.7

Table 4. Level 02 & Above Live loads

Room Occupancy or Use	Uniform (kN/m2)
1. Service Lift Lobby	5
2. Probe Areas	15
3. Electrical Room	10
4. Mechanical Room	10
5. All General Areas	5

Table 5. Roof Level Super Imposed Dead Loads

Room Occupancy or Use	Uniform (kN/m2)
6. Staircase/Lift Roof	4.7
7. All General Roof Areas	4.7
8. All General Roof Service Areas	4.7

Table 6. Roof Level Live loads

Room Occupancy or Use	Uniform (kN/m2)
1. Staircase/Lift Roof	1.5
2. All General Roof Areas	5.0
3. All General Roof Service Areas	7.5

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8.0 Document Control

6.1 Approval: GLOBAL_FAC_GFTT_APPR

6.2 Notification:

GLOBAL_FAC_MANAGERS
GLOBAL_FAC_COMMUNICATIONS
GLOBAL_FAC_CONSTRUCTION
GLOBAL_FAC_NOTIFY
FCG_F4_FAC_NOTIFY
FCG_F6_FAC_NOTIFY
FCG_F10_FAC_NOTIFY
FCG_F11_FAC_NOTIFY
FCG_F15_FAC_NOTIFY
FCG_F16_FAC_NOTIFY
FCG_MMP_FAC_NOTIFY
FCG_MMY_FAC_NOTIFY
FCG_MSB_FAC_NOTIFY
FCG_MXA_FAC_NOTIFY
FCG_MTB_FAC_NOTIFY

6.3 Notification:

All Micron Team Members at all Sites

6.4 Retention

Adherence to the requirements specified in the
[Global Facilities - Document Control Procedures](#)

6.5 Review

Biennially (two years)

9.0 Revision History

Revision #	Section Changed	Details of Changes	Date
0	NA	New Document	06/02/2022
1	Section 5	Table 1 and Table 2 Additional SDL and LL for WWT Loft Station and UPW Tank	10/26/2022
	Section 7	Table 4 and Table 6 Revised LL for Probe Area and LL for All General Roof Areas	